

MiniGenius – Rewards System, Parent Dashboard, Navigation & State Management

1. نظام Rewards (Stickers + Achievements + Daily Gift) 2. Parent Dashboard 3. Navigation & Routing Structure 4. State Management اختيارية (Provider) 5. ملاحظات Production Build

1. Rewards System

1.1 نموذج الملصق (Sticker Model)

```
class Sticker {
    final String id;
    final String name;
    final String assetPath;
    bool unlocked;

    Sticker({
        required this.id,
        required this.name,
        required this.assetPath,
        this.unlocked = false,
    });
}
```

1.2 StickerManager

```
class StickerManager {
    final List<Sticker> stickers = [
        Sticker(id: 'happy_brain', name: 'Happy Brain', assetPath: 'assets/stickers/happy_brain.png'),
        Sticker(id: 'winner_brain', name: 'Winner Brain', assetPath: 'assets/stickers/winner_brain.png'),
        Sticker(id: 'star_badge', name: 'Star Badge', assetPath: 'assets/stickers/star_badge.png'),
    ];

    void unlockSticker(String id) {
        final sticker = stickers.firstWhere((s) => s.id == id, orElse: () => throw Exception('Sticker not found'));
        sticker.unlocked = true;
    }
}
```

```

}

List<Sticker> get unlocked => stickers.where((s) => s.unlocked).toList();
List<Sticker> get locked => stickers.where((s) => !s.unlocked).toList();
}

```

شاشة ألبوم الملصقات 1.3

```

class StickerAlbumScreen extends StatelessWidget {
  final StickerManager manager;

  const StickerAlbumScreen({super.key, required this.manager});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Sticker Album')),
      body: GridView.builder(
        padding: const EdgeInsets.all(16),
        gridDelegate: const SliverGridDelegateWithFixedCrossAxisCount(
          crossAxisCount: 3,
          crossAxisSpacing: 12,
          mainAxisSpacing: 12,
        ),
        itemCount: manager.stickers.length,
        itemBuilder: (context, index) {
          final sticker = manager.stickers[index];
          return Opacity(
            opacity: sticker.unlocked ? 1 : 0.3,
            child: Container(
              decoration: BoxDecoration(
                color: Colors.white,
                borderRadius: BorderRadius.circular(16),
              ),
              child: Column(
                mainAxisAlignment: MainAxisAlignment.center,
                children: [
                  Icon(Icons.emoji_emotions, size: 40), // استبدالها بصورة الملصق
                  const SizedBox(height: 8),
                  Text(sticker.name, style: const TextStyle(fontSize: 12)),
                ],
              ),
            ),
          );
        },
      ),
    );
  }
}

```

```
    );  
  }  
}
```

1.4 Achievements Model

```
class Achievement {  
  final String id;  
  final String title;  
  final String description;  
  bool unlocked;  
  
  Achievement({  
    required this.id,  
    required this.title,  
    required this.description,  
    this.unlocked = false,  
  });  
}
```

1.5 AchievementManager

```
class AchievementManager {  
  final List<Achievement> achievements = [  
    Achievement(id: 'level_5', title: 'Level Explorer', description: 'Finish 5  
levels'),  
    Achievement(id: 'memory_master', title: 'Memory Master', description:  
'Finish all memory levels'),  
  ];  
  
  void unlock(String id) {  
    final ach = achievements.firstWhere((a) => a.id == id, orElse: () => throw  
Exception('Not found'));  
    ach.unlocked = true;  
  }  
  
  List<Achievement> get unlocked => achievements.where((a) =>  
a.unlocked).toList();  
}
```

1.6 Daily Gift Logic (Pseudo)

- SharedPreferences يستخدم تاريخ اليوم من
- Daily Gift إذا آخر دخول مختلف عن اليوم الحالي → عرض نافذة

```
class DailyGiftService {
    final SharedPreferences prefs;
    static const String lastClaimKey = 'last_daily_claim';

    DailyGiftService(this.prefs);

    bool canClaimToday() {
        final last = prefs.getString(lastClaimKey);
        final today = DateTime.now().toIso8601String().substring(0, 10);
        return last != today;
    }

    void claimToday() {
        final today = DateTime.now().toIso8601String().substring(0, 10);
        prefs.setString(lastClaimKey, today);
    }
}
```

2. Parent Dashboard

2.1 نموذج بيانات اللعب

```
class PlaySession {
    final DateTime date;
    final int minutes;
    final String gameId;

    PlaySession({required this.date, required this.minutes, required
    this.gameId});
}

class ParentStats {
    final List<PlaySession> sessions;

    ParentStats(this.sessions);
}
```

```

    int get totalMinutes => sessions.fold(0, (sum, s) => sum + s.minutes);
}

```

2.2 مبسطة Parent Dashboard شاشة

```

class ParentDashboardScreen extends StatelessWidget {
  final ParentStats stats;

  const ParentDashboardScreen({super.key, required this.stats});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Parent Dashboard')),
      body: Padding(
        padding: const EdgeInsets.all(16.0),
        child: Column(
          crossAxisAlignment: CrossAxisAlignment.start,
          children: [
            Text('Total Play Time: ${stats.totalMinutes} minutes',
              style: const TextStyle(fontSize: 18)),
            const SizedBox(height: 16),
            const Text('Recent Sessions:', style: TextStyle(fontSize: 16)),
            const SizedBox(height: 8),
            Expanded(
              child: ListView.builder(
                itemCount: stats.sessions.length,
                itemBuilder: (context, index) {
                  final s = stats.sessions[index];
                  return ListTile(
                    leading: const Icon(Icons.videogame_asset),
                    title: Text('Game: ${s.gameId}'),
                    subtitle: Text('${s.minutes} min on ${s.date.toLocal()}'),
                  );
                },
              ),
            ),
          ],
        ),
      ),
    );
  }
}

```

3. Navigation & Routing Structure

3.1 Routes Constants

```
class AppRoutes {  
  static const splash = '/';  
  static const home = '/home';  
  static const memory = '/memory';  
  static const shape = '/shape';  
  static const pattern = '/pattern';  
  static const stickers = '/stickers';  
  static const achievements = '/achievements';  
  static const parentDashboard = '/parent-dashboard';  
}
```

3.2 MaterialApp with Routes

```
class MiniGeniusApp extends StatelessWidget {  
  const MiniGeniusApp({super.key});  
  
  @override  
  Widget build(BuildContext context) {  
    return MaterialApp(  
      debugShowCheckedModeBanner: false,  
      initialRoute: AppRoutes.splash,  
      routes: {  
        AppRoutes.splash: (_) => const SplashScreen(),  
        AppRoutes.home: (_) => const HomeScreen(),  
        AppRoutes.memory: (_) => MemoryGameScreen(),  
        AppRoutes.shape: (_) => ShapeMatcherScreen(),  
        AppRoutes.pattern: (_) => PatternLogicScreen(),  
        AppRoutes.stickers: (_) => StickerAlbumScreen(manager:  
StickerManager()),  
        AppRoutes.parentDashboard: (_) => ParentDashboardScreen(  
          stats: ParentStats([]),  
        ),  
      },  
    );  
  }  
}
```

4. State Management (Provider)

4.1 AppState كمثل جامع

```
class AppState extends ChangeNotifier {  
  final StickerManager stickerManager = StickerManager();  
  final AchievementManager achievementManager = AchievementManager();  
  
  int totalScore = 0;  
  
  void addScore(int value) {  
    totalScore += value;  
    notifyListeners();  
  }  
}
```

4.2 في أعلى الشجرة Provider ربط

```
void main() {  
  runApp(  
    ChangeNotifierProvider(  
      create: (_) => AppState(),  
      child: const MiniGeniusApp(),  
    ),  
  );  
}
```

4.3 داخل لعبة AppState استخدام

```
// داخل MemoryGameScreen  
onTap: () {  
  final flipped = controller.flipCard(index);  
  if (flipped) {  
    context.read<AppState>().addScore(1);  
  }  
}
```

5. Production Build Notes

- **release mode:** flutter build apk --release
 - التأكد من:
 - flutter_launcher_icons إضافة الأيقونات عبر
 - إن احتجت ProGuard تفعيل
 - الحقيقية AdMob IDs ربط
 - اختبار على عدة شاشات وأجهزة
 - ربطها في Privacy Policy إعدادات Play Store
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END OF FILE

وجاهز Rewards, Parent Dashboard, Navigation, State Management, هذا الملف يكمل MiniGenius من ناحية Production. للتحويل لبروجيكت.